

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester IV)
Dot .Net Core and Progressive Web Application Practical

Practical – VII

Roll No.: T044	Name: SACHIN SINGH
Class: TYIT	Batch: 1
Date of Assignment: 04/08/2026	Date/Time of Submission: 04/08/2026

Aim(A) Develop a PWA that works in offline mode using Service Worker.

CODE :

index.cshtml :

```
@{
    ViewData["Title"] = "Collage Portal";
}
<div class="text-center">
    <h1 class="display-4">Collage Website</h1>
    <h2>Time Table Download</h2>
    <h2>Notices Download</h2>
    <h2>Study Material Download</h2>
</div>
```

_Layout.cshtml :

```
<link rel="manifest" href="/manifest.json" />
```

[Before Head closing add this part]

```
<script>
    if ('serviceWorker' in navigator) {
        navigator.serviceWorker.register('/service-worker.js');
    }
</script>
```

[Before Body closing add this part]

service-worker.js

```
self.addEventListener("install", event => {
    console.log("Service Worker Installed");
});
self.addEventListener("fetch", event => {
    event.respondWith(
        fetch(event.request)
    );
});
```

manifest.json :

```
{
    "name": "College Information Portal",
    "short_name": "College Portal",
    "start_url": "/",
    "display": "standalone",
    "icons": [
        {
            "src": "icons/icon1.png",
```

```

    "sizes": "128x128",
    "type": "image/png"
  },
  {
    "src": "icons/icon2.png",
    "sizes": "512x512",
    "type": "image/png"
  }
]
}

```

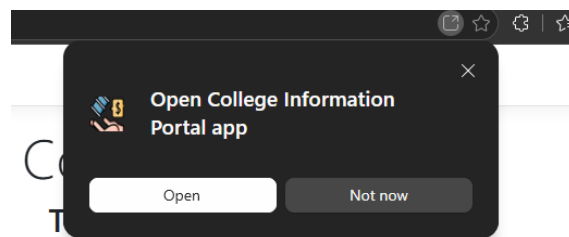
OUTPUT :

Home Privacy

Offline Page Student Record

Roll No : 44

Name : SACHIN



Notices Download

Offline Page Code Changes :

service-worker.js :

```

const CACHE = "v1";

self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/offlinePage.html"
      ]);
    })
  );
});

console.log("Service Worker Installed");
});

self.addEventListener("fetch", e => {
  e.respondWith(
    fetch(e.request).catch(() =>
      caches.match("/offlinePage.html")
    )
  );
});

```

OfflinePage.html :

```

<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>Offline</title>
</head>

<body>

```

```

<h1>Welcome to offline College Portal </h1>

<p>Student Details </p>
<p>Name : Harshita Kanojia </p>
<p>Roll.No. : T011 </p>
<p>Department : Information Techonlogy </p>
<hr />
<h1>College Dashboard</h1>
<p>Time Table Download</p>
<p>Notices Download</p>
<p>Study Material Download</p>
<hr />
<p>Please check your internet connection and try again.</p>

</body>
</html>

```

OUTPUT :

Home Privacy

Offline Page Student Record

Roll No : 44

Name : SACHIN

Aim(B) : Create a Student Record Management PWA.

Additional Code :

Student.cs :

```

namespace CuroosityQuestion.Models
{
    public class Student
    {
        public int RollNo { get; set; }
        public string Name { get; set; }
    }
}

```

Index.cshtml :

```

@{
    ViewData["Title"] = "Collage Portal";
}
<div class="text-center">
    <h1 class="display-4">Collage Website</h1>
    <h1>Offline Page</h1>
    <h2>Student Record</h2>
    <p>Roll No : @Model.RollNo</p>
    <p>Name : @Model.Name</p>

```

</div>

service-worker :

```
const CACHE = "v1";

self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/Home/Index",
        "/css/site.css",
        "/js/site.js"
      ]);
    })
  );
});

self.addEventListener("fetch", e => {
  e.respondWith(
    caches.match(e.request).then(response => {
      return response || fetch(e.request);
    })
  );
});
```

HomeController.cs :

```
Student s = new Student();
s.RollNo = 11;
s.Name = "Harshita";

return View(s);
```

[add after public IActionResult Index()]

OUTPUT :

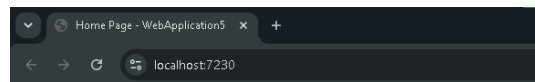
Home Privacy

Offline Page

Student Record

Roll No : 44

Name : SACHIN

[WebApplication5](#)

- [Home](#)
- [Privacy](#)

Offline Page**Student Record**

Roll No : 44

Name : SACHIN

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